

GROTLI III, Norway

WMO: 013610

Lat: 62.016N

Lon: 7.664E

Elev: 872

StdP: 91.28

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.8	(c) -26.4	(d) -31.8	(e) 0.2	(f) -28.8	(g) -29.3	(h) 0.3	(i) -26.4	(j) 11.3	(k) -5.7	(l) 10.3	(m) -6.2	(n) 0.4	(o) 340	(p) 0.769

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	20.9	12.6	18.9	11.8	16.9	11.1	13.6	18.8	12.7	17.5	11.8	16.0	3.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.3	9.2	15.3	10.2	8.6	14.5	9.4	8.1	13.9	40.9	18.7	38.3	17.7	35.9	16.0	17.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.8	7.6	6.7	-31.7	23.3	4.0	2.3	-34.6	24.9	-36.9	26.2	-39.2	27.5	-42.1	29.1
			-31.9	14.6	4.0	1.4	-34.7	15.6	-37.0	16.4	-39.3	17.2	-42.1	18.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-9.5	-8.4	-5.5	-1.4	3.5	7.4	10.9	9.6	6.4	0.5	-4.8	-8.0
	DBStd	8.72	7.35	6.77	5.69	4.79	3.84	2.89	3.12	2.76	3.23	4.14	5.97	7.43
	HDD10.0	3717	605	516	481	343	204	86	27	41	115	296	443	559
	HDD18.3	6657	863	749	739	593	461	327	231	269	359	554	693	817
	CDD10.0	103	0	0	0	0	2	9	54	30	6	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	11	0	0	0	0	0	0	11	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	3.1	3.6	3.3	3.7	2.9	2.9	3.1	2.7	2.7	3.0	3.2	3.0	3.1
Precipitation	PrecAvg	989	105	93	98	48	46	66	67	72	88	96	96	112
	PrecMax	1349	276	185	342	181	104	150	108	162	267	217	261	212
	PrecMin	627	0	0	15	4	11	12	22	10	23	17	2	16
	PrecStd	184	78	55	75	36	21	32	24	34	58	47	78	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	5.2	5.7	10.4	19.8	20.9	25.2	20.3	17.7	13.5	7.3	5.5
		MCWB	1.7	2.1	2.6	5.0	11.4	11.6	15.1	12.1	10.3	9.4	4.6	3.3
	2%	DB	2.6	3.1	4.3	8.1	15.7	18.4	22.3	18.6	15.5	10.0	5.4	3.5
		MCWB	1.0	0.9	2.0	4.0	8.8	10.7	13.4	11.6	10.2	6.7	3.3	1.8
	5%	DB	1.3	1.6	3.1	6.4	12.7	15.7	20.1	16.8	13.7	7.9	3.8	2.1
		MCWB	-0.2	-0.3	1.0	2.8	7.4	9.1	12.9	11.3	9.4	5.4	2.1	0.5
	10%	DB	-0.3	0.1	1.7	5.0	10.6	13.9	17.9	15.1	12.0	6.1	2.2	0.6
		MCWB	-1.6	-1.4	-0.2	1.9	6.2	8.9	12.0	10.6	8.6	3.5	0.7	-0.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	2.7	3.3	5.6	11.5	12.9	15.6	13.7	12.4	10.8	4.9	3.5
		MCDB	3.6	4.8	5.0	9.4	18.7	18.0	22.7	17.4	15.6	13.8	7.1	5.2
	2%	WB	1.2	1.0	2.2	4.2	9.2	11.6	14.4	12.9	10.7	6.9	3.6	1.9
		MCDB	2.5	2.8	4.1	7.4	14.4	16.7	20.4	16.9	13.9	9.3	5.4	3.5
	5%	WB	-0.1	-0.2	1.2	3.1	7.7	10.3	13.4	12.0	9.8	5.4	2.1	0.6
		MCDB	1.2	1.4	3.0	6.2	12.5	14.8	18.9	15.7	13.2	7.9	3.7	1.9
	10%	WB	-1.6	-1.5	0.0	2.0	6.3	9.0	12.5	11.1	8.7	3.8	0.8	-0.8
		MCDB	-0.4	0.1	1.5	4.6	10.3	13.2	17.4	14.5	12.0	5.9	2.1	0.4
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.2	8.7	9.2	8.8	9.4	9.7	8.9	8.1	7.0	6.9	7.8
		MCDBR	7.1	6.5	7.2	9.4	13.0	14.5	15.1	11.9	11.7	9.3	6.7	6.5
		MCWBR	6.0	5.2	5.4	5.8	7.6	8.0	7.5	6.5	7.1	6.5	5.3	5.6
	5% WB	MCDBR	7.0	6.5	7.0	8.4	13.0	12.3	13.1	10.3	10.3	7.8	6.7	6.7
		MCWBR	6.1	5.3	5.5	5.2	7.8	7.1	6.9	6.0	6.7	5.8	5.3	5.8
Clear-Sky Solar Irradiance	taub	0.204	0.254	0.288	0.319	0.324	0.323	0.334	0.324	0.295	0.272	0.210	0.177	
	taud	2.035	2.184	2.158	2.138	2.203	2.316	2.383	2.411	2.473	2.348	2.062	1.918	
	Ebn at Noon	501	720	819	863	892	901	880	859	826	698	478	328	
	Edh at Noon	35	64	96	121	124	113	103	92	71	54	33	21	
All-Sky Solar Radiation	RadAvg	0.19	0.70	1.89	3.57	4.71	4.87	4.59	3.52	2.13	0.97	0.28	0.11	
	RadStd	0.04	0.17	0.33	0.46	0.47	0.48	0.44	0.34	0.28	0.14	0.06	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)	N/A	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page